

The record costs recoil: a floor for every probe state

For every protocol of momentum-transfer marks, with probes in any states whatever (Gaussian or not, squeezed, correlated, mixed, grid states), deciding between free motion and a constant force F over a duration τ at error probability ϵ requires

$$s \sum_j \Delta_j \geq 8(1 - 2\epsilon)\hbar, \quad s = \frac{F\tau^2}{2m},$$

where s is Newton's sagitta for the interval and Δ_j is the spread of the impulse delivered by mark j . The resolutions of the marks do not enter. For a general force the sagitta is replaced by the uniform distance of the forced displacement from straight motion (Theorem R).

The theorem answers the question the non-Gaussian probes raise. The floor $\tau\Delta E \geq 24z^2\hbar\sqrt{(1-\rho)/(1+\rho)}$ of the mark-cost note is a theorem about Gaussian probes: a grid state with zero position-momentum correlation decides the comparison at any force (Proposition G), so no bound on second moments alone survives. What does survive, for every probe, is the recoil. A pointer registers a displacement only through the spread of its conjugate momentum, and that momentum is the impulse the body receives. This is M3 in its universal form: a record of the sagitta must leave the delivered impulse undetermined by at least $8(1 - 2\epsilon)\hbar/s$ in total. The correlated three-mark protocol of Proposition D attains the bound within a factor $\sqrt{2}z_{1-\epsilon}/(1 - 2\epsilon)$, uniformly in its squeezing (Section 4). Exploratory; no ledger promotion.

1. Setting

Transverse coordinate $\hat{y}$, momentum $\hat{p}$, mass m , on $[0, \tau]$. Marks $j = 1, \dots, k$ act at times t_j : the body is coupled to probe j , prepared in a state φ_j (pure or mixed, arbitrary), by $U_j = \exp(-\frac{i}{\hbar}\lambda_j\hat{y}\hat{P}_j)$, and the pointer $\hat{Q}_j$ is read afterwards with no further evolution. Under hypothesis I the body moves freely; under F it feels a force whose displacement relative to free motion from rest at the origin is $P(t)$, with $P(0) = P'(0) = 0$; for the constant force $P(t) = Ft^2/2m$. The body's initial state is unknown: a test **decides at error** ϵ when its equal-prior error probability is at most ϵ for every initial state under I and every initial state under F. Write $\hat{N}_j = \hat{Q}_j/\lambda_j$ and $\hat{D}_j = -\lambda_j\hat{P}_j$ for the error and the delivered impulse of mark j , as in Theorem A, and $\Delta_j = \lambda_j\Delta_{\varphi_j}\hat{P}_j$ for its recoil spread.

Lemma 1 (the record). In the Heisenberg picture the j -th reading is

$$\hat{R}_j = \hat{y} + \frac{\hat{p}t_j}{m} + \theta P(t_j) + \sum_{i: t_i < t_j} \frac{t_j - t_i}{m} \hat{D}_i + \hat{N}_j,$$

with $\theta = 0$ under I and $\theta = 1$ under F, and the initial pointer $\hat{Q}_j$ appears in $\hat{R}_j$ alone.

Proof. Theorem A of the mark-cost note for each coupling, the linear free evolution between marks, and the c-number force, which adds $P(t)$ to $\hat{y}(t)$. The pointer $\hat{Q}_j$ is shifted by the coupling and coupled to nothing afterwards. $\square$

Lemma 2 (the signal moves into the pointers). For real a, b let the body's initial state under F be the state under I translated by $-a$ in position and $-mb$ in momentum. Then the record under F has the distribution of the record under I with each probe state φ_j replaced by its translate by $\lambda_j c_j$ in $\hat{Q}_j$, where $c_j = P(t_j) - a - bt_j$.

Proof. The translation of the body replaces $\hat{y}, \hat{p}$ in Lemma 1 by $\hat{y} - a, \hat{p} - mb$, which shifts $\hat{R}_j$ by $-a - bt_j$; the force adds $P(t_j)$. The net effect is the c-number shift $\hat{R}_j \mapsto \hat{R}_j + c_j$. Translating probe j by $\lambda_j c_j$ in $\hat{Q}_j$ shifts $\hat{N}_j$ by c_j and leaves every $\hat{D}_i$ and every other reading unchanged, by the last clause of Lemma 1. The record operators commute, so their joint distribution is fixed by these shifts and the states. $\square$

2. The theorem

Theorem R. Every protocol deciding at error ϵ satisfies

$$1 - 2\epsilon \leq \frac{1}{\hbar} \min_{a,b} \sum_j \Delta_j |P(t_j) - a - bt_j|.$$

Proof. Fix a, b and any initial state σ under I, and take under F the translate of Lemma 2. A test with error at most ϵ against this pair has $\text{TV} \geq 1 - 2\epsilon$ between the two record distributions. By Lemma 2 those are the records of one fixed protocol, with body state σ , applied to the probe states $\otimes_j \varphi_j$ and $\otimes_j \varphi'_j$, φ'_j the translate by $q_j = \lambda_j c_j$. The data-processing inequality and the triangle inequality over the factors give

$$\text{TV} \leq \frac{1}{2} \|\otimes_j \varphi_j - \otimes_j \varphi'_j\|_1 \leq \sum_j \frac{1}{2} \|\varphi_j - \varphi'_j\|_1.$$

The translate is $e^{-iq_j \hat{P}_j / \hbar} \varphi_j e^{iq_j \hat{P}_j / \hbar}$. By Mandelstam–Tamm for a pure state, and through a purification for a mixed one exactly as in Theorem 2 of the probabilistic note, $\frac{1}{2} \|\varphi_j - \varphi'_j\|_1 \leq |q_j| \Delta_{\varphi_j} \hat{P}_j / \hbar = \Delta_j |c_j| / \hbar$. Minimize over a, b . $\square$

Corollary R1 (the Galileo comparison). For $P(t) = Ft^2/2m$, the line $a + bt$ that best approximates P uniformly on $[0, \tau]$ is $\frac{F}{2m}(\tau t - \tau^2/8)$, and $|P(t) - a - bt| \leq F\tau^2/16m = s/8$ there, with equality at $t = 0, \tau/2, \tau$. Hence

$$s \sum_j \Delta_j \geq 8(1 - 2\epsilon)\hbar.$$

For a general force the same step gives $E_1(P) \sum_j \Delta_j \geq (1 - 2\epsilon)\hbar$, with $E_1(P) = \min_{a,b} \max_{[0,\tau]} |P(t) - a - bt|$ the uniform distance of the forced displacement from straight motion. $\square$

The mechanism is the conjugacy of Theorem A read in the other direction. The record of the force is a translation of the pointers, and a pointer feels a translation only through the spread of the operator that generates it, which is its momentum $\hat{P}_j$. That same momentum, times λ_j , is the impulse the body receives. So sensitivity is paid in recoil at the exchange rate $\hbar$, whatever the pointer's state, and a resolution bought by squeezing, correlation or a grid is bought with recoil.

For adaptive protocols, in which couplings and times depend on earlier readings, the hybrid argument of the probabilistic note gives the same bound with each $\Delta_j |c_j|$ replaced by its supremum over the choices available at step j .

3. The Gaussian floor needs Gaussian probes

Proposition G. In the three-mark protocol of Proposition D of the mark-cost note (marks at $0, \tau/2, \tau$, test $R_1 - 2R_2 + R_3$, outer marks sharp), let the middle probe be a grid state with a real wavefunction, so that $\text{Cov}(\hat{Q}_2, \hat{P}_2) = 0$ and its correlation ρ vanishes. For every F whose signal $u^\top P = F\tau^2/4m$ is not a multiple of $\sigma = \sqrt{2\pi\hbar|u_2 S_2|/m} = \sqrt{2\pi\hbar\tau/m}$, the error probability tends to zero as the grid's teeth are narrowed at fixed spacing.

Proof. The statistic's noise is $-2\hat{N}_2 + \frac{\tau}{2m}\hat{D}_2$ plus the vanishing contributions of the outer marks, that is $\alpha\hat{Q}_2 + \beta\hat{P}_2$ with $\alpha\beta = |u_2 S_2|/m = \tau/m$ independent of λ_2 . Take the ideal grid state stabilized by the commuting translations $e^{-i\ell_Q \hat{P}_2 / \hbar}$ and $e^{i\ell_P \hat{Q}_2 / \hbar}$, $\ell_Q \ell_P = 2\pi\hbar$ (Gottesman, Kitaev and Preskill, PRA **64**, 012310, 2001, metadata), with the aspect chosen as $\ell_P / \ell_Q = \alpha / \beta$. The product of the two stabilizers is $-e^{ic(\alpha\hat{Q}_2 + \beta\hat{P}_2)}$ with $c = \ell_P / (\hbar\alpha)$, by the Baker–Campbell–Hausdorff phase $e^{-i\pi}$, so $\alpha\hat{Q}_2 + \beta\hat{P}_2$ is supported on a comb of spacing $2\pi/c = \sqrt{2\pi\hbar\alpha\beta} = \sigma$. Finite-energy grid states with Gaussian teeth of width w and a Gaussian envelope keep a real

wavefunction and converge to the ideal state as $w \rightarrow 0$; the statistic then has teeth of width of order w , and a shift by a non-multiple of σ separates the two combs. $\square$

Such states measure both quadratures of a small displacement at once, which is the principle of the grid-state displacement sensor of Duivenvoorden, Terhal and Weigand (PRA **95**, 012305, 2017, metadata). Two features connect it to the rest of the programme. Its recoil diverges: the envelope of the momentum distribution widens as $\hbar/w$, so Theorem R is met. And its blind spots are the shifts in $\sigma\mathbb{Z}$, the translations whose Weyl phase with the stabilizers is a multiple of 2π : the grid reads the signal modulo the same central phase that the polygon-lift note identifies as the part no preparation can move.

So Theorems B and C of the mark-cost note hold for Gaussian probes, where the Wigner function is a density and the record is the classical Gaussian model, and the correlation coefficient is the resource that lowers the floor within that class. Outside it the second moments do not bound the error, and Theorem R is the statement that holds.

4. Tightness

In the correlated protocol of Proposition D the outer marks are sharp, so their recoils are unbounded and the minimum in Theorem R takes the line through the endpoints, $c_1 = c_3 = 0$, $|c_2| = F\tau^2/8m$. The balanced middle mark has $2\delta_2 = \tau\Delta_2/2m$ and, for a pure Gaussian probe, $\delta_2\Delta_2 = \hbar/(2\sqrt{1-\rho^2})$, so $\Delta_2^2 = 2m\hbar/(\tau\sqrt{1-\rho^2})$. At its decision threshold $\tau\Delta E = 32z^2\hbar\sqrt{(1-\rho)/(1+\rho)}$,

$$\Delta_2|c_2| = z_{1-\epsilon}\hbar\sqrt{\frac{2}{1+\rho}},$$

between $z\hbar$ and $\sqrt{2}z\hbar$ for every $\rho \in [0,1)$. Theorem R asks for $(1-2\epsilon)\hbar$, so the protocol sits within the factor $\sqrt{2}z_{1-\epsilon}/(1-2\epsilon)$ of the bound, about 2.6 at five per cent error, however strongly it is squeezed. The recoil-weighted sagitta is the right universal resource: squeezing moves the protocol along it and never below it.

5. What this changes

The three floors, ordered. Resource-free: the phase $\mathcal{K}_\tau/\hbar$ between the motion and its inscribed polygon (polygon-lift note, Theorems 1 and 7). Universal over probe states: the recoil bound $s\sum_j\Delta_j \geq 8(1-2\epsilon)\hbar$ (Theorem R). Universal over instruments with a bounded laboratory: the aperture bound $F\tau L + \frac{F\tau^2}{2m}P \geq (1-2\epsilon)\hbar$ (probabilistic note). And for Gaussian probes with correlation at most ρ , the sharp floor $\tau\Delta E \geq 24z^2\hbar\sqrt{(1-\rho)/(1+\rho)}$ (mark-cost note).

M3, restated. The mark-floor note's premise has two clauses, a product of resolution and recoil and the independence of the two. Theorem R shows that the recoil clause alone is universal: a mark that records the sagitta must leave the delivered impulse undetermined, with $s\sum\Delta_j \geq 8(1-2\epsilon)\hbar$. In Newton's words from the mark-floor note, the premise that carries $h > 0$ for every probe is that a record costs an undetermined impulse, and the pairing is with the sagitta of Lemma X.

Beyond the momentum-transfer class. The proof uses only that the pointer's error is translated by the unitary generated by the delivered impulse, $[\hat{N}_j, \hat{D}_j] = -i\hbar$, and that the apparatus enters the record through $\hat{N}_j$ and $\hat{D}_j$. The commutation of the pointer with the body's momentum after any coupling gives $[\hat{N}, \hat{D}] = -i\hbar - [\hat{y}, \hat{D}] - [\hat{N}, \hat{p}]$, so Theorem R extends to every instrument whose error and recoil do not depend on the body (Arthurs and Goodman, PRL **60**, 2447, 1988, metadata). That is the next step, and the body-dependent case, where Ozawa's relation (PRA **67**, 042105, 2003, metadata) adds the body's own spreads, is the one after.

6. Consequence for STATE

Theorem A's extension, option 3 of the 2026-09-22 plan, is settled constructively. The Gaussian floor holds for Gaussian probes and grid probes break it (Proposition G); the universal statement over probe states is Theorem R, sagitta times total recoil at least $8(1 - 2\epsilon)\hbar$, attained within a constant by squeezed probes at every squeezing. Next: option 1 (additive-noise instruments), then option 2 (Ozawa's relation).